

MODULE IV: Statistical Techniques II

1. Introduction to Probability

Definition: Probability measures how likely an event is to occur.

$$P(E) = \frac{\text{Favourable outcomes}}{\text{Total outcomes}}$$

Properties: 1. $0 \leq P(E) \leq 1$ 2. $P(S) = 1$ (Certain event) 3. $P(\phi) = 0$ (Impossible event) 4. $P(\text{not } E) = 1 - P(E)$

Example: A coin tossed once: $P(H) = 1/2$, $P(T) = 1/2$

2. Laws of Probability

(a) Addition Law

Used when events are **either/or** type.

For **mutually exclusive** events:

$$P(A \cup B) = P(A) + P(B)$$

For **non-mutually exclusive** events:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Example: Probability of drawing King or Heart in a deck of cards:

$$P(K \cup H) = \frac{4}{52} + \frac{13}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}$$

(b) Multiplication Law

Used when events are **both/and** type.

$$P(A \cap B) = P(A) \times P(B|A)$$

If A and B are **independent**, then $P(A \cap B) = P(A)P(B)$

Example: Two heads in two coin tosses: $P = 1/2 \times 1/2 = 1/4$

3. Conditional Probability

Definition: Probability of event A given that event B has already occurred.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Example: Probability that a drawn card is King given it is a face card:

$$P(K|F) = \frac{4/52}{12/52} = 1/3$$

4. Bayes' Theorem

Used to find the probability of causes given the result.

$$P(A_i|B) = \frac{P(A_i)P(B|A_i)}{\sum P(A_j)P(B|A_j)}$$

Example: Two urns: Urn1 (4W,6B), Urn2 (4W,5B). One chosen randomly, white drawn.

$$P(U_1|W) = \frac{(1/2)(4/10)}{(1/2)(4/10) + (1/2)(4/9)} = 9/19$$

$$P(U_2|W) = 10/19$$

5. Random Variables

Definition: A random variable assigns numerical values to outcomes of a random experiment.

- **Discrete random variable:** Takes countable, distinct numerical values such as 0, 1, 2, 3, ... It can be represented using a **probability mass function (PMF)**, which gives the probability for each possible value of X.

Formula:

$$P(X = x_i) = p_i, \text{ where } \sum p_i = 1$$

Example: Number of heads in 3 coin tosses $\rightarrow X = \{0, 1, 2, 3\}$

$$P(X=k) = (3 \text{ choose } k)(1/2)^3$$

Mean: $E(X) = \sum x_i P(X=x_i)$

Variance: $\text{Var}(X) = \sum (x_i - E(X))^2 P(X=x_i)$

- **Continuous random variable:** Takes any real value in a continuous interval or range (for example, $0 \leq x \leq 10$). It is represented using a **probability density function (PDF)**.

Formula:

$$P(a < X < b) = \int_a^b f(x) dx, \text{ where } \int_{-\infty}^{+\infty} f(x) dx = 1$$

Example: Height of students or time taken to complete a task.

If $X \sim \text{Uniform}(0,1)$, then $f(x)=1$ for $0 \leq x \leq 1$.

Mean: $E(X) = \int_{-\infty}^{+\infty} x f(x) dx$

Variance: $\text{Var}(X) = \int_{-\infty}^{+\infty} (x - E(X))^2 f(x) dx$